

Nd-induced enhancement of magnetic damping in L1₀-FePt for HAMR

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In heat-assisted magnetic recording (HAMR), recording is performed during the cooling process of media from its Curie temperature. Completing magnetization switching within a few hundred picoseconds during this cooling period, with a sufficiently high magnetic damping constant, is essential to improve the switching probability and overall recording performance. This requirement becomes increasingly critical at higher areal densities with steeper thermal gradients. Rare-earth Nd doping in L1₀-FePt enables modulation of magnetic properties including the anisotropy field and Curie temperature, as experimentally confirmed in this study. In addition, our results show that Nd doping leads to an enhancement of the magnetic damping constant at elevated temperatures relevant to HAMR recording. These modulations of magnetic properties and magnetic damping constant are suggested to originate from an increase in spin-orbit coupling induced by the introduction of localized rare-earth 4f states. These results demonstrate the feasibility of designing HAMR media by tuning the magnetic damping constant through rare-earth element doping.

Index Terms—Heat-assisted magnetic recording, Nd doping, magnetic damping.

I. INTRODUCTION

MAGNETIZATION dynamics at elevated temperatures are crucial for recording in heat-assisted magnetic recording (HAMR). In HAMR, the media are heated above the Curie temperature (T_C), and the magnetization is aligned along the head magnetic field during cooling. This process occurs between T_C and the freezing temperature (T_f), below which the anisotropy field ($\mu_0 H_k$) exceeds the applied head magnetic field, thereby defining the recording time window [1]. Completing magnetization switching within the recording time window on the order of hundreds of picoseconds and allowing the magnetization to reach thermal equilibrium by T_f , are necessary to suppress back-switching and improve switching probability and recording performance such as bit error rate [2], [3]. With increasing areal density, the recording time window shortens due to steeper thermal gradients, requiring faster magnetization switching [1], [3].

Media with a higher damping constant (α) are effective in improving switching probability and recording performance due to faster magnetization switching [3]. Previous studies have shown that rare-earth element doping is effective in increasing α in NiFe because of enhanced spin-orbit coupling (SOC) through localized rare-earth 4f states [4]. While the modification of SOC-related properties by rare-earth doping is also theoretically predicted for L1₀-FePt, which is utilized as the recording layer material in HAMR media, its impact on α has not been experimentally clarified yet [5].

In this study, we investigated Nd-doped L1₀-FePt to clarify the effects of rare-earth doping on microstructure, static magnetic properties at room temperature, and magnetization dynamics at elevated temperatures.

II. EXPERIMENTS

Samples of (FePt)_{100-x}(Nd)_x ($x = 0.0 - 4.1$ at.%) with thicknesses ranging from 30 to 50 nm were deposited on MgO (001) single-crystal substrates by sputtering at a substrate temperature of 500 °C. A carbon capping layer was deposited on (FePt)_{100-x}(Nd)_x layer to avoid oxidation. The Nd composition was analyzed by x-ray fluorescence (XRF).

The microstructure and Nd distribution were evaluated using scanning transmission electron microscopy (STEM), energy-dispersive x-ray spectroscopy (EDS), and atom probe tomography (APT). The L1₀-ordering was analyzed from the superlattice (001) and (003) peaks and the fundamental (002) peaks of L1₀-FePt measured by x-ray diffraction (XRD). Static magnetic properties, such as $\mu_0 H_k$ and T_C , were measured using MPMS3, SQUID-VSM and PPMS Dynacool.

The magnetization dynamics were investigated by all-optical time-resolved magneto-optical Kerr effect (TRMOKE) with a pump-probe method. The external magnetic field (H_{ext}) was applied using a superconducting magnet at a fixed angle of 80° from the film normal. To measure magnetization dynamics at elevated temperature, the sample in a vacuum chamber was heated using the heater embedded in sample holder, and its temperature was monitored by an attached thermocouple.

For each temperature condition from room temperature to around T_C , magnetization dynamics were measured under several H_{ext} conditions in the range of 5-7 T, and frequency of magnetization precession (f) and lifetime of magnetization precession (τ) were obtained to estimate $\mu_0 H_k$, and α .

III. RESULTS AND DISCUSSIONS

According to XRD, superlattice peaks of L1₀-FePt were observed in all (FePt)_{100-x}(Nd)_x samples ($x = 0.0 - 4.1$ at.%). The L1₀-ordering decreased with increasing Nd content from 0.87 to 0.75. In contrast, $\mu_0 H_k$ and T_C initially increased with Nd

content, reaching maximum values at Nd contents of 0.7 and 1.5 at.%, respectively, and then decreased at higher Nd contents. The reduction in $\mu_0 H_k$ and T_C at higher Nd contents can be primarily attributed to the reduced L1₀-ordering. The initial enhancement of $\mu_0 H_k$ is consistent with the theoretical predictions for rare-earth-doped L1₀-FePt, in which modification of SOC-related properties has been suggested [5].

Fig. 1 shows the magnetization dynamics of (FePt)_{100-x}(Nd)_x samples ($x = 0.0, 4.1$ at.%) measured at 643 K under an H_{ext} of 7 T. For each condition, the effective damping constant (α_{eff}) was calculated using Eq. (1).

$$\frac{1}{\tau} = 2\pi\alpha_{\text{eff}}f \quad (1)$$

Fig. 2 shows the temperature dependence of α_{eff} at H_{ext} of 7 T. Temperature is normalized by T_C , which was determined by fitting the temperature dependence of the magnetization saturation using the Kuz'min formula [6]. α_{eff} initially decreased with increasing temperature, and then increased around T_C . A similar trend was observed in nanogranular L1₀-FePt [7]. α_{eff} includes both intrinsic Gilbert damping and extrinsic contributions. Extrinsic contributions originating from magnetic inhomogeneities such as the distribution of $\mu_0 H_k$ are suppressed at elevated temperatures where the absolute value of $\mu_0 H_k$ becomes smaller. In contrast, intrinsic Gilbert damping increases at high temperature due to thermally driven spin fluctuation. Around T_C , where intrinsic damping is expected to dominate and magnetization switching in HAMR is triggered, samples with higher Nd contents exhibited larger α_{eff} from 0.048 for the undoped sample to 0.066 for the Nd-doped sample (4.1 at.%).

The observed enhancement of α_{eff} suggests a modification of SOC induced by Nd doping, which is also supported by the enhancement of $\mu_0 H_k$ and T_C at low Nd contents. In rare-earth-doped NiFe, the enhancement of magnetic damping via SOC has been reported [4]. Such an enhancement is often discussed in terms of localized 4f-related SOC effects, and a similar mechanism may contribute to the enhancement of damping in Nd-doped L1₀-FePt.

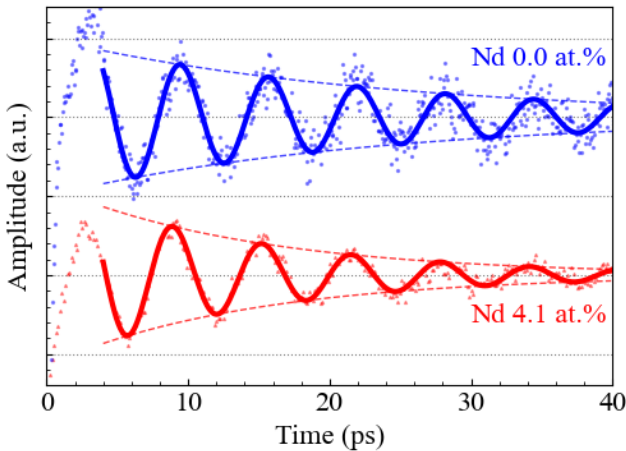


Fig. 1. Magnetization dynamics of (FePt)_{100-x}(Nd)_x samples ($x = 0.0, 4.1$ at.%) measured at 643 K under $H_{\text{ext}} = 7$ T.

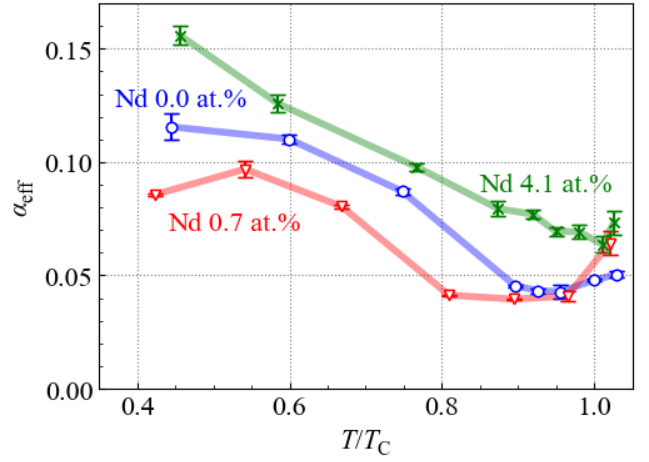


Fig. 2. Temperature dependence of α_{eff} under $H_{\text{ext}} = 7$ T, where the temperature is normalized by T_C . The values of T_C for Nd contents of 0.0, 0.7, and 4.1 at.% were obtained from the temperature dependence of the saturation magnetization and were 674, 709, and 657 K, respectively.

IV. SUMMARY

Nd-doped L1₀-FePt was investigated with respect to microstructure, static magnetic profile, and magnetization dynamics. The modulation of $\mu_0 H_k$ and T_C by Nd doping implies modulation of SOC in addition to L1₀-ordering reduction.

Temperature dependence of α_{eff} is influenced by both intrinsic Gilbert damping and extrinsic contributions. Around T_C where HAMR recording occurs and also Gilbert damping is expected to dominate α_{eff} , α_{eff} increased from 0.048 to 0.066 for Nd 4.1 at.%. These results demonstrate the feasibility of designing α of HAMR media.

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